OWASP Serverless:7 Severity: Critical Status: Fixed ✓

IAM Blast Radius via Over-Privileged Lambda Role

ICS-344 Information Security · KFUPM · DVSA Project

OWASP Category: Serverless:7 – Over-Privileged Function Permissions
Affected Function: DVSA-SEND-RECEIPT-EMAIL
Severity: Critical
Status: Fixed and Verified

1. Goal and Vulnerability Summary

The goal of this lesson is to demonstrate how an over-privileged IAM execution role attached to a Lambda function dramatically increases the blast radius of any compromise. The **DVSA-SEND-RECEIPT-EMAIL** function exists solely to send a receipt email after an order is placed. Its role, however, grants unrestricted read/write access to every S3 bucket and every DynamoDB table in the account, as well as full SES administrative permissions.

By temporarily modifying the function to print its environment variables — simulating what an attacker could achieve through code injection or a vulnerable dependency — the temporary STS credentials can be extracted and used locally to impersonate the function's identity and perform any AWS action the role permits, including scanning the entire customer orders database.

Impact: Complete data breach. Any attacker who achieves code execution inside the function immediately inherits account-wide S3, DynamoDB, and SES permissions — far beyond what the function needs to send a single receipt email.

2. Why This Works / Root Cause

In AWS serverless systems, every Lambda function runs with an IAM execution role. The function's temporary credentials (issued by STS) are automatically injected into the Lambda runtime environment as `AWS_ACCESS_KEY_ID`, `AWS_SECRET_ACCESS_KEY`, and `AWS_SESSION_TOKEN`. Any code running inside the function — including injected code — can read and use these credentials.

The root cause is **over-privileged role design**. The attached policies used wildcard resources:

```
"Resource": ["arn:aws:s3:::*", "arn:aws:s3:::*/*"] "Resource":
["arn:aws:dynamodb:us-east-1:*:table/*", "arn:aws:dynamodb:us-east-1:*:table/*/index/*"]
```

This means the role has permission to read, write, and delete objects in **any** S3 bucket and scan, modify, or delete items in **any** DynamoDB table — not just the DVSA receipts resources. The function also had `AmazonSESFullAccess` attached.

Root cause: Failure to enforce the Principle of Least Privilege on the Lambda execution role.

3. Environment and Setup

Component	Detail
Platform	AWS Lambda (Python 3.8)
Vulnerable Function	DVSA-SEND-RECEIPT-EMAIL
Execution Role	serverlessrepo-OWASP-DVSA-SendReceiptFunctionRole-i6gMbFqAvx5E
Over-Privileged Policies	SendReceiptFunctionRolePolicy1 (S3), SendReceiptFunctionRolePolicy2 (DynamoDB), AmazonSESFullAccess
Evidence Tool	IAM Policy Simulator (policysim.aws.amazon.com)

Exploit Tool	AWS CLI with stolen STS credentials
Target Table	DVSA-ORDERS-DB
AWS Region	us-east-1

4. Reproduction Steps

Step 1 — Identify the execution role

Navigate to Lambda → DVSA-SEND-RECEIPT-EMAIL → Configuration → Permissions and open the execution role in IAM. Review the attached policies.

Step 2 — Confirm over-privilege in policy JSON

Open SendReceiptFunctionRolePolicy1 and observe the wildcard S3 resources. Open SendReceiptFunctionRolePolicy2 and observe the wildcard DynamoDB table resources.

Step 3 — Confirm with IAM Policy Simulator

Select the role and run s3:GetObject and dynamodb:Scan against arbitrary non-DVSA resources. Both return **Allowed**, confirming the blast radius extends beyond the function's scope.

Step 4 — Expose credentials via environment variable print

Add the following line to send_receipt_email.py immediately after the handler definition, then deploy:

```
print(dict(os.environ))
```

Step 5 — Trigger the function and extract credentials from CloudWatch

Place a new order on the DVSA website. Navigate to CloudWatch → /aws/lambda/DVSA-SEND-RECEIPT-EMAIL → most recent log stream and extract AWS_ACCESS_KEY_ID, AWS_SECRET_ACCESS_KEY, and AWS_SESSION_TOKEN.

Step 6 — Assume the function's identity locally

```
export AWS_ACCESS_KEY_ID="<value from logs>" export AWS_SECRET_ACCESS_KEY="<value from logs>"
export AWS_SESSION_TOKEN="<value from logs>"
```

Step 7 — Exploit the over-privileged role

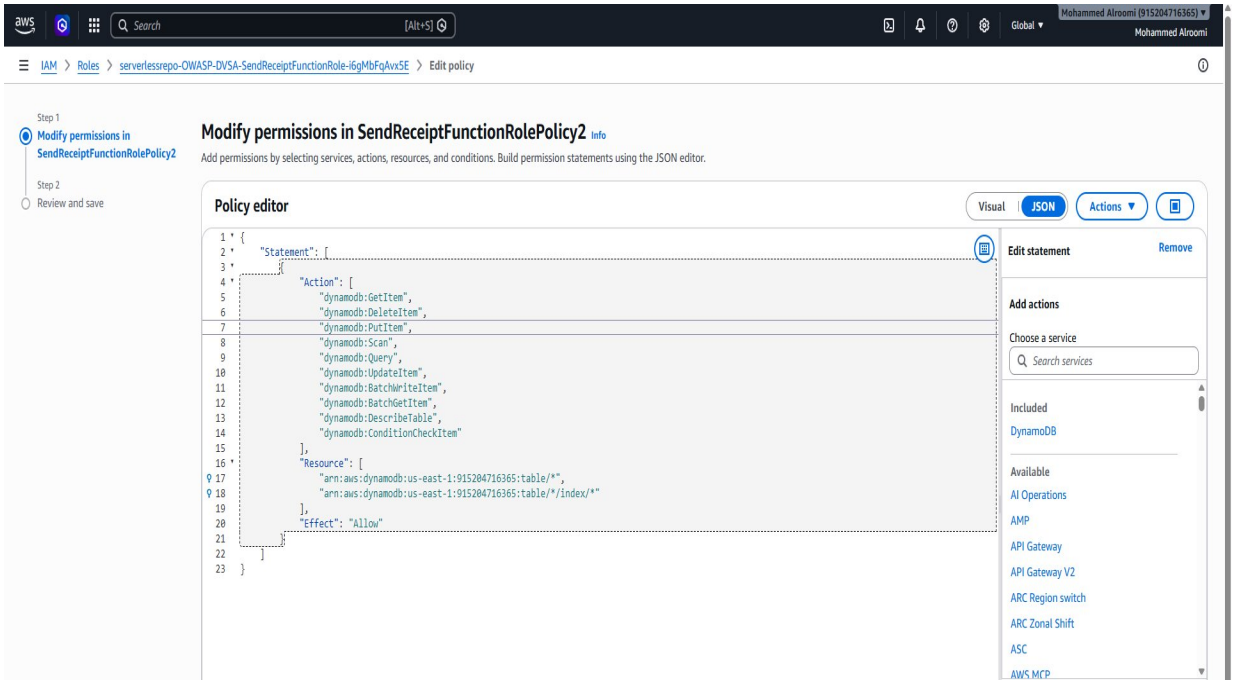
```
aws dynamodb scan --table-name DVSA-ORDERS-DB --region us-east-1
```

The entire customer orders database is returned, demonstrating a complete data breach.

5. Evidence and Proof

DynamoDB wildcard policy (SendReceiptFunctionRolePolicy2)

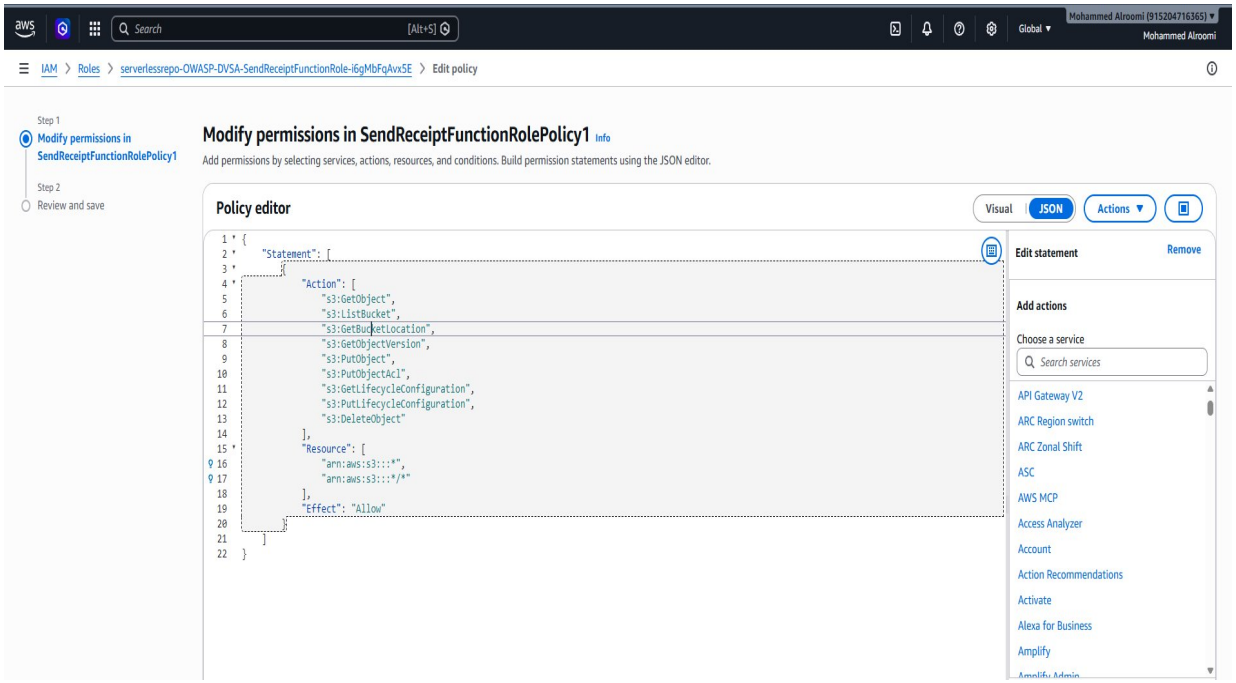
The policy grants full DynamoDB access across all tables in the account using table/* wildcards.



SendReceiptFunctionRolePolicy2 showing table/ wildcard resources*

S3 wildcard policy (SendReceiptFunctionRolePolicy1)

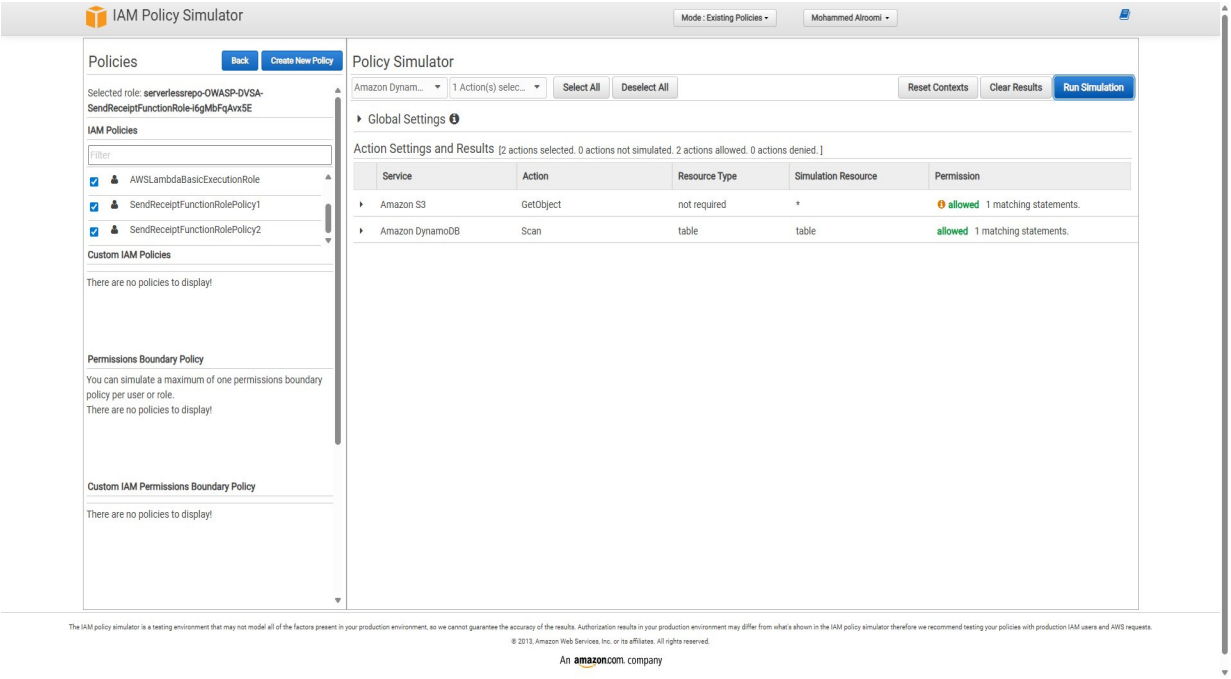
The policy grants S3 actions across all buckets and all objects — far beyond the single DVSA receipts bucket.



SendReceiptFunctionRolePolicy1 showing arn:aws:s3::: wildcard resources*

IAM Policy Simulator — both actions Allowed

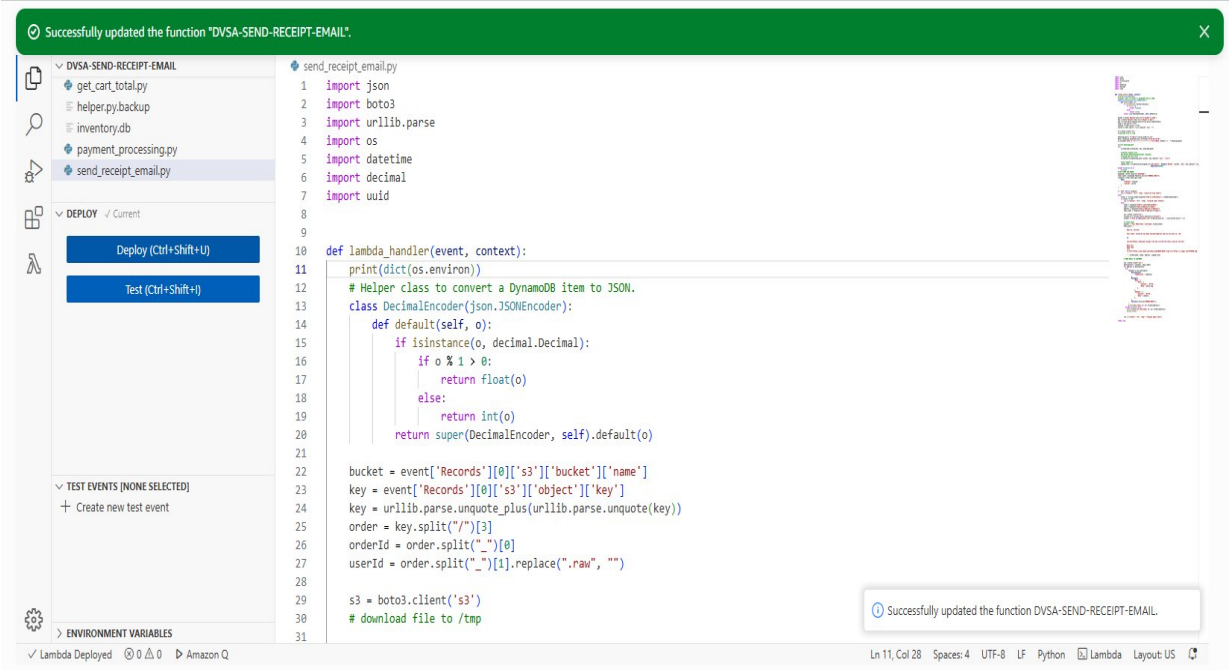
s3:GetObject and dynamodb:Scan are both Allowed for non-DVSA resources, confirming the blast radius extends beyond the function's intended scope.



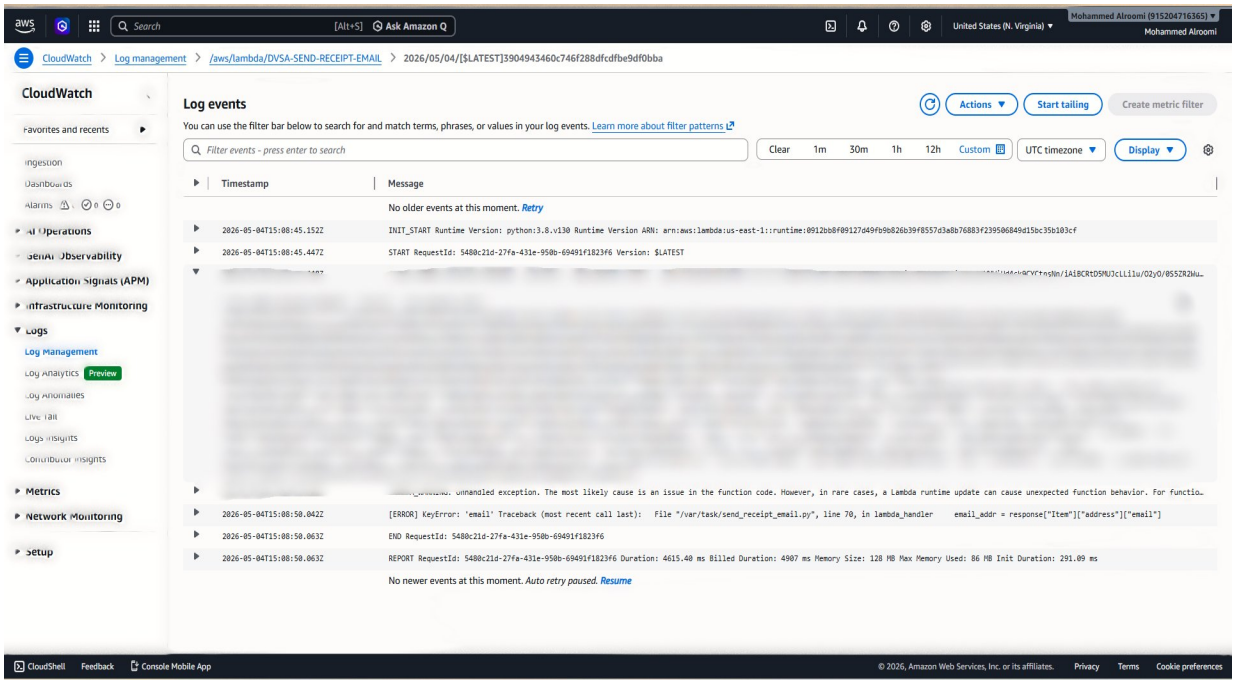
IAM Policy Simulator confirming Allowed for S3 and DynamoDB

Credentials exposed in CloudWatch

The print(dict(os.environ)) line causes the Lambda runtime to log all environment variables including the temporary STS credentials.



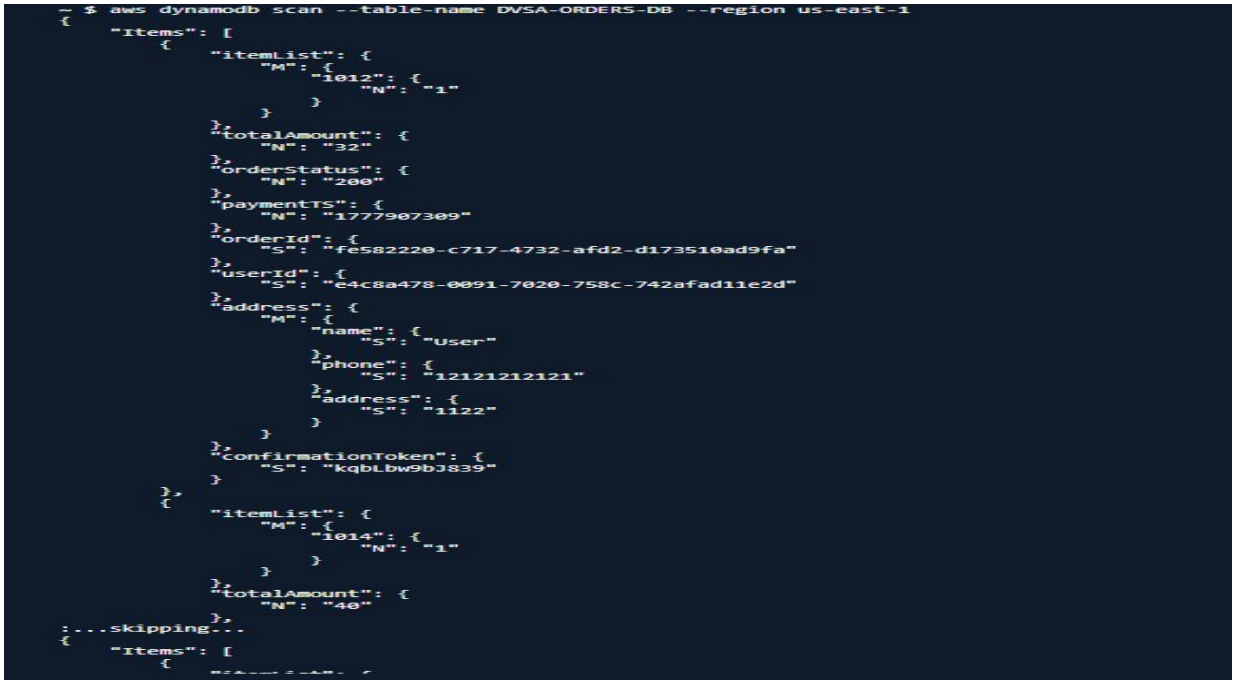
Lambda code editor showing the print(dict(os.environ)) line deployed



CloudWatch log showing environment dictionary with credentials (redacted)

■ Exploit proof — full orders database dumped

Using the stolen credentials locally, aws dynamodb scan returns every customer order in DVSA-ORDERS-DB — demonstrating a complete data breach using only the stolen Lambda credentials.



Terminal: aws dynamodb scan dumping customer orders database (1/3)

```
            {
                "itemList": {
                    "M": {
                        "1012": {
                            "N": "1"
                        }
                    }
                },
    {
        "Items": [
            {
                "itemList": {
                    "M": {
                        "1012": {
                            "N": "1"
                        }
                    }
                },
                "totalAmount": {
                    "N": "32"
                },
                "orderStatus": {
                    "N": "200"
                },
                "paymentTS": {
                    "N": "1777907309"
                },
                "orderId": {
                    "S": "fe582220-c717-4732-afd2-d173510ad9fa"
                },
                "userId": {
                    "S": "e4c8a478-0091-7020-758c-742afad11e2d"
                },
                "address": {
                    "M": {
                        "name": {
                            "S": "User"
                        },
                        "phone": {
                            "S": "12121212121"
                        },
                        "address": {
                            "S": "1122"
                        }
                    }
                },
                "confirmationToken": {
                    "S": "kqbLbw9bJ839"
                }
            },
            {
                "itemList": {
                    "M": {
                        "1014": {
                            "N": "1"
                        }
                    }
                },
```


Terminal: continued dump of customer orders (2/3)


```
                    }
                },
                "totalAmount": {
                    "N": "40"
                },
                "orderStatus": {
                    "N": "300"
                },
                "paymentTS": {
                    "N": "1777058470"
                },
                "orderId": {
    :
```


Terminal: final entries in the orders database dump (3/3)

6. Fix Strategy / Probable Mitigation

The fix applies the **Principle of Least Privilege** to the execution role:

- Remove SendReceiptFunctionRolePolicy1 (wildcard S3 access)
- Remove SendReceiptFunctionRolePolicy2 (wildcard DynamoDB access)
- Remove AmazonSESFullAccess (full SES admin)
- Add a new inline policy LeastPrivilegeReceiptPolicy scoped to: s3:GetObject / s3:PutObject on the DVSA receipts bucket only; dynamodb:GetItem on DVSA-ORDERS-DB only; ses:SendEmail and ses:SendRawEmail
- Remove the print(dict(os.environ)) line from the function code

7. Code / Config Changes

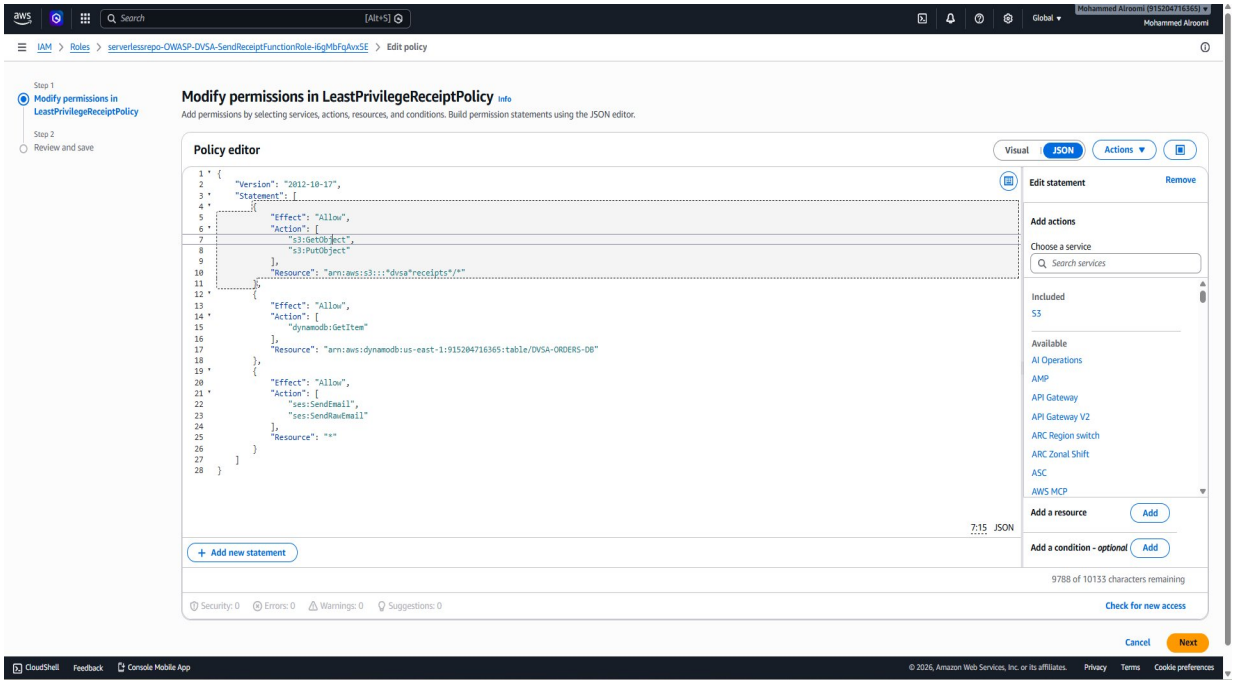
Vulnerable policies (before)

```
// SendReceiptFunctionRolePolicy1 - S3 wildcard "Action":
["s3:GetObject","s3:ListBucket","s3:PutObject","s3:DeleteObject",...] "Resource":
["arn:aws:s3:::*", "arn:aws:s3:::*/*"]
```

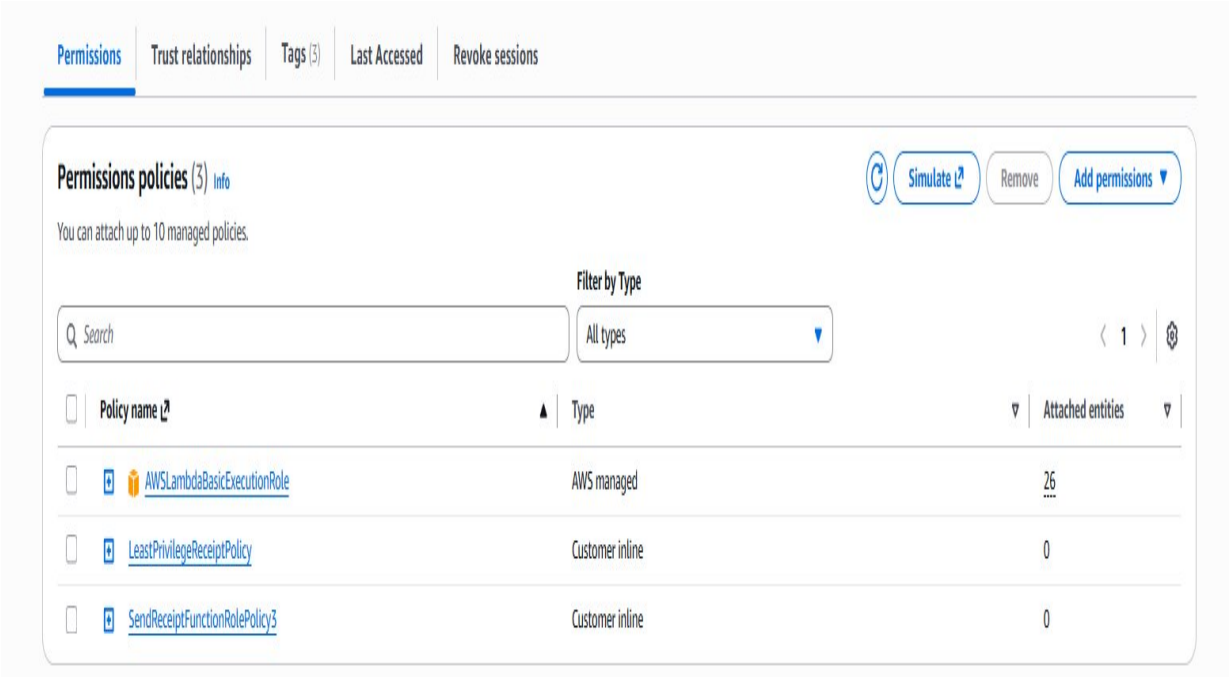
```
// SendReceiptFunctionRolePolicy2 - DynamoDB wildcard "Action":
["dynamodb:GetItem","dynamodb:Scan","dynamodb:PutItem","dynamodb:DeleteItem",...] "Resource":
["arn:aws:dynamodb:us-east-1:915204716365:table/*",
"arn:aws:dynamodb:us-east-1:915204716365:table/*/index/*"]
```

Fixed policy (after) — LeastPrivilegeReceiptPolicy

```
{ "Version": "2012-10-17", "Statement": [ { "Effect":"Allow",
"Action":["s3:GetObject","s3:PutObject"], "Resource":"arn:aws:s3:::*dvsa*receipts*/*" }, {
"Effect":"Allow", "Action":["dynamodb:GetItem"],
"Resource":"arn:aws:dynamodb:us-east-1:915204716365:table/DVSA-ORDERS-DB" }, {
"Effect":"Allow", "Action":["ses:SendEmail","ses:SendRawEmail"], "Resource":"*" } ] }
```



LeastPrivilegeReceiptPolicy JSON in IAM policy editor

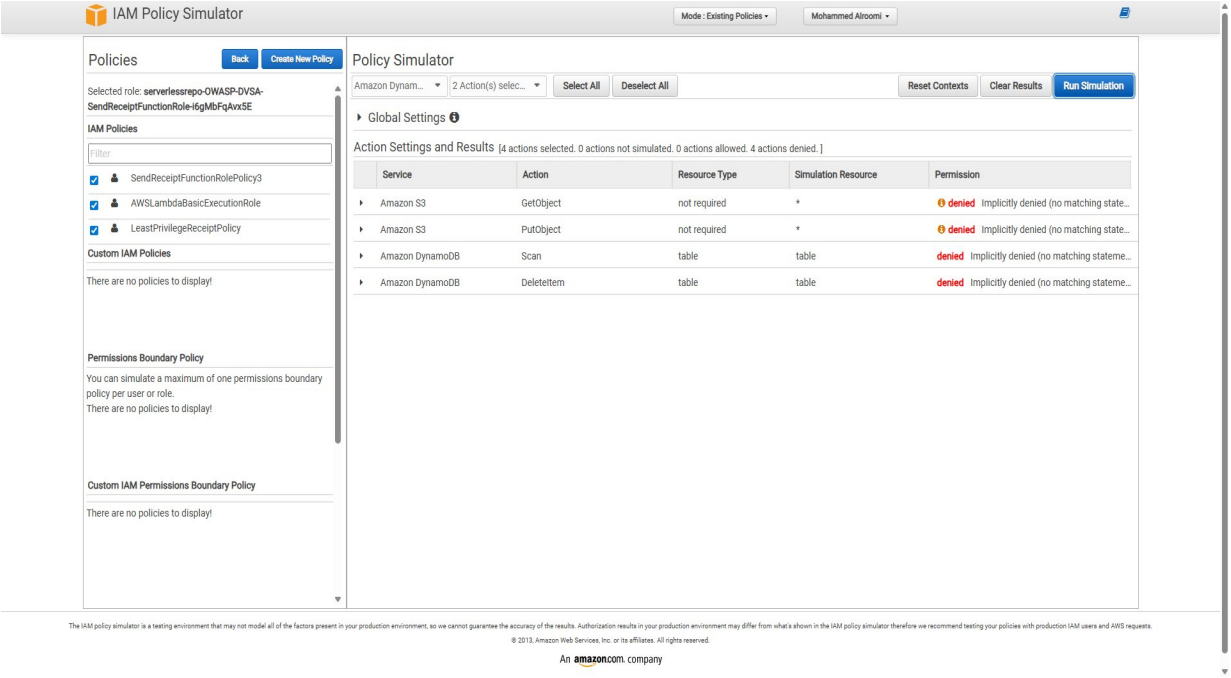


IAM role showing the new LeastPrivilegeReceiptPolicy replacing the old wildcard policies

8. Verification After Fix

Policy Simulator — actions now Denied

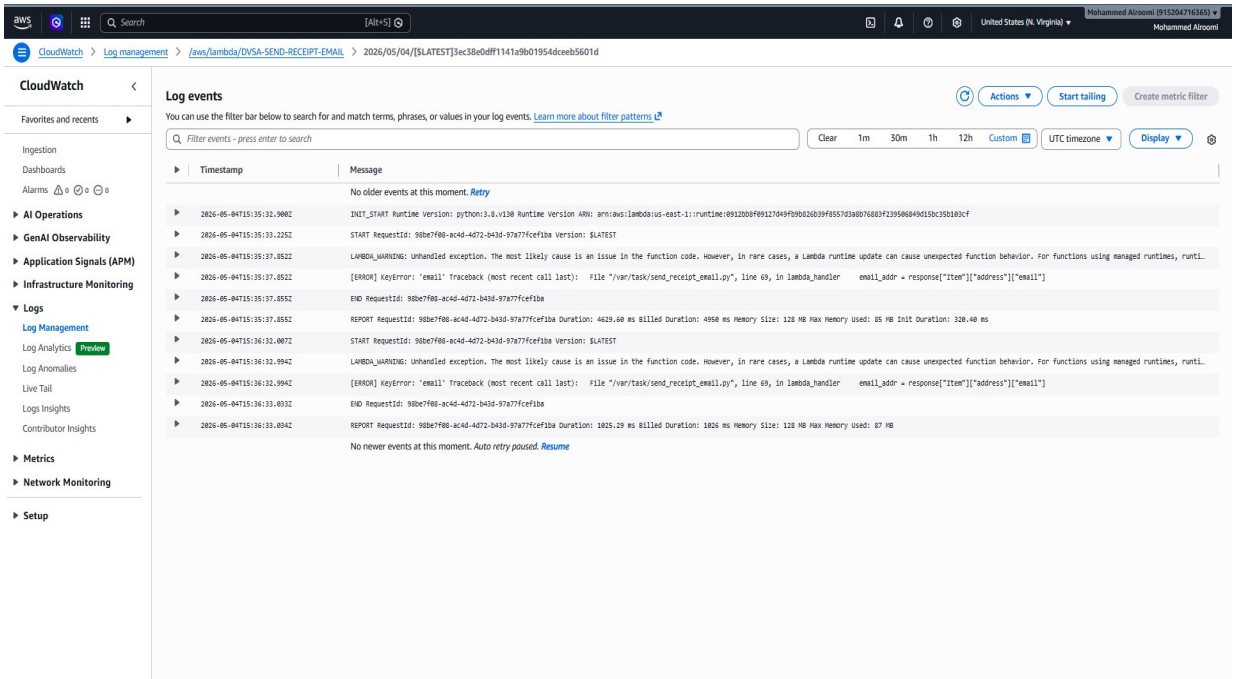
After applying LeastPrivilegeReceiptPolicy and removing the wildcard policies, re-running the same simulation confirms that s3:GetObject, s3:PutObject, dynamodb:Scan, and dynamodb:DeleteItem on arbitrary non-DVSA resources are now Denied.



IAM Policy Simulator showing all 4 actions Denied after the fix

Function still operates correctly

A new order placed after the fix triggers DVSA-SEND-RECEIPT-EMAIL successfully. CloudWatch confirms a clean invocation with no credentials printed and no errors related to the permission change.



CloudWatch log showing clean function invocation after fix — no credentials printed

9. Structured Operation and Security Analysis

Table A — Structured Analysis

Vulnerability	Intended Rule(s)	Artifacts Used to Infer Rule	Normal Behavior Evidence	Exploit Behavior Evidence

Over-Privileged Function	DVSA-SEND-RECEIPT-EMAIL must only access the DVSA receipts S3 bucket, DVSA-ORDERS-DB, and SES send actions. It must not access any other AWS resource.	IAM role permissions tab; SendReceiptFunctionRole Policy1 & Policy2 JSON; IAM Policy Simulator results; CloudWatch logs; AWS CLI scan output.	Function invoked after an order → sends receipt email → terminates. Only interacts with the specific order in S3 and DynamoDB.	After extracting STS credentials from CloudWatch logs, aws dynamodb scan returned all customer orders — far beyond the function's intended scope.

Table B — Deviation and Fix Analysis

Vulnerability	Why This Is a Deviation	Deviation Class	Fix Applied (Where)	Post-Fix Verification	Optional Latency Before/After
Over-Privileged Function	The execution role granted wildcard S3 and DynamoDB permissions allowing access to every bucket and table in the account — far beyond receipt sending scope.	Accidental misconfiguration	Replaced Policy1, Policy2, AmazonSESFullAccess with LeastPrivilegeReceipt Policy scoped to the receipts bucket, DVSA-ORDERS-DB, and minimal SES actions.	IAM Policy Simulator confirms S3 and DynamoDB wildcard actions are now Denied. Function continues to invoke successfully.	Not measured

10. Takeaway / Lessons Learned

1. **In serverless, the function role is the security perimeter.** Unlike a traditional server where an attacker must escalate privileges separately, a Lambda function's temporary credentials are immediately available to any code running inside it. Over-privilege converts any code execution flaw into an account-level breach.

2. **Blast radius is a design decision.** The difference between a contained incident and a full data breach is determined at role design time — before any vulnerability is found. Wildcard resources expand blast radius unnecessarily.

3. **Least Privilege must be resource-scoped, not just action-scoped.** Allowing only dynamodb:GetItem is not enough if it applies to table/*. The resource ARN must be pinned to the exact table, bucket, and prefix the function needs.

4. **CloudTrail and IAM Access Analyzer help right-size roles.** AWS provides tooling to determine the minimum permissions a function actually uses. There is no justification for not using these tools during deployment.

5. **The fix is architectural, not cosmetic.** Removing wildcard policies and replacing them with scoped ones does not change the function's legitimate behavior — but it ensures that any future compromise cannot pivot to unrelated resources.